

Reception Dashboard

Logic & Calculation Document

LabWise LIMS

8 April 2026

Stats Strip (Top Bar)

Processed Today

How many samples this receptionist processed (received into the system) today.

Logic: Count Sample records where receivedBy = this user AND createdAt = today.

Received Today

How many samples this receptionist received that are still in "received" status today.

Logic: Count Sample records where receivedBy = this user, status = "received", AND createdAt = today.

Pending TRFs

How many TRFs are globally waiting to be booked. This is a system-wide number — every receptionist sees the same count.

Logic: Count all Sample records where status = "received" (no user filter).

Avg Time

Average time in minutes between a sample arriving and the receptionist logging it.

Logic: For each sample today: (Sample.createdAt - Sample.receivedDate) in minutes. Average all values. If receivedDate is missing, createdAt is used (resulting in 0 minutes).

Approval Rate

What percentage of this month's samples have moved past "received" status.

Logic: (samples not in "received" status / total samples this month) x 100. Defaults to 100% if no samples.

All stats refresh every 60 seconds automatically.

KPIs (5 Metrics)

All KPIs auto-computed from database. No manual input needed.

KPI	Target	What It Measures
TRFs Processed / Day	30	Volume of samples registered today
Avg Processing Time	10 mins	Time from sample arrival to system entry
Approval Accuracy	99%	Percentage of samples moving past received status
Pending TRFs	0	System-wide queue of unbooked samples
New Client Activations	Info only	New clients onboarded this month

Detailed Calculations

1. TRFs Processed / Day

Count of Sample records where receivedBy = this user AND createdAt = today.

Target: 30. Color: Green >= 90% of target, Amber >= 75%, Red < 75%.

2. Avg Processing Time

- For each sample today: (Sample.createdAt - Sample.receivedDate) in minutes
 - Average all values, rounded to 1 decimal
 - Fallback: if receivedDate is null, uses createdAt (= 0 minutes)
- Target: 10 minutes. Inverse KPI — lower is better. Progress = target / actual.

3. Approval Accuracy

Samples that moved past "received" / total processed this month x 100.

If a sample stays stuck in "received" it counts against accuracy.

Target: 99%.

4. Pending TRFs

Global count of samples in "received" status. Not user-scoped — shows lab-wide queue health.

Target: 0. Inverse KPI — lower is better.

5. New Client Activations

Unique clients from this month's samples whose Client.createdAt is also this month.

Informational only — no target, no color scoring.

KRAs (4 Weighted Areas)

Overall KRA = weighted average of 4 scores. Each gets a score out of 100.

KRA	Target	Weight	How Score is Calculated
Processing Accuracy	99%	30%	Intact samples / total samples x 100
Processing Speed	95%	25%	% of samples processed within 10 mins
Client Verification	90%	25%	% of samples with valid client assigned
Doc Compliance	100%	20%	% of samples with all required fields filled

Overall KRA Formula

(Accuracy Score x 0.30) + (Speed Score x 0.25) + (Verification Score x 0.25) + (Compliance Score x 0.20)

Example:

- Processing Accuracy: 97.5% actual, target 99% => Score: $\min(100, 97.5/99 \times 100) = 98.5\%$
 - Processing Speed: 92% actual, target 95% => Score: $\min(100, 92/95 \times 100) = 96.8\%$
 - Client Verification: 100% actual, target 90% => Score: $\min(100, 100/90 \times 100) = 100\%$ (capped)
 - Doc Compliance: 95.5% actual, target 100% => Score: $\min(100, 95.5/100 \times 100) = 95.5\%$
- Overall = $(98.5 \times 0.30) + (96.8 \times 0.25) + (100 \times 0.25) + (95.5 \times 0.20) = 29.6 + 24.2 + 25.0 + 19.1 = 97.9\%$

Color Coding:

- Green: Score $\geq 90\%$
- Yellow: Score $\geq 75\%$
- Red: Score $< 75\%$

TRF Queue Logic

Sorting

- First by priority: Express > Urgent > Normal
- Within same priority: earliest received first (receivedDate ascending)
- Limited to 30 records

Wait Time Colors

Wait Time	Color	Meaning
≤ 30 minutes	Green	Timely processing
31 - 60 minutes	Yellow	Getting slow
> 60 minutes	Red	Needs immediate attention

New Client Detection

If a sample's client has no client code (code is null/empty), it is flagged with a purple "New Client" badge. This tells the receptionist to trigger client activation before approving.

Alerts (Auto-Triggered)

Alert Type	Trigger Condition	Severity	Refresh
Urgent/Express TRFs	Sample in received status with priority urgent or express	Red	30s
Damaged Samples	Sample in received status with condition damaged/ temperature deviation/leakinn	Amber	30s
Long-Waiting TRFs	Sample in received status with receivedDate > 2 hours ago	Amber	30s

Auto-Refresh Schedule

Data	Refresh	Stale Time	Reason
Alerts + TRF Queue	30 seconds	15s	Samples arrive frequently, need fast response
Stats + Received + Activity	60 seconds	30s	Work volume changes during the day
KPIs + KRAs	5 minutes	60s	Monthly metrics change slowly

If server is unavailable, dashboard shows mock data so the page doesn't break.

Data Sources

What	Database Table
Sample counts, status, dates, condition	Sample
Client names, codes, new client detection	Client
Product type names	ProductType
User identity for scoping	User (via JWT)

Pending Items

Item	Current State	What Will Change
Sample Condition KRA	Treats non-intact as receptionist issue	Separate arrival condition from processing errors
Client Complaints KPI	Defined in role config but not on dashboard	Add complaint tracking mechanism
Booking Accuracy KPI	Dashboard uses Approval Accuracy instead	Align with role config or update config
Configurable Time Target	10 min hardcoded	Make configurable per location/role
Location-Scoped Pending	Pending TRFs is global count	Filter by receptionist's location